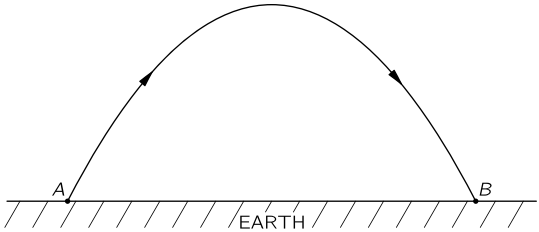

Figure 6-19 In a uniform gravitational field the trajectory with the maximum proper time for a fixed elapsed time is a parabola.

time than clock *B*. You see that we have a kind of game. If we stand still with clock *A* we get 100 seconds. If we go up slowly to a small height and come down slowly we can get a little more than 100 seconds. If we go a little higher, maybe we can gain a little more. But if we go too high we have to move fast to get there, and we may slow down the clock enough that we end up with less than 100 seconds. What program of height versus time—how high to go and with what speed to get there, carefully adjusted to bring us back to clock *B* when it has increased by 100 seconds—will give us the largest possible time reading on clock *A*?

Answer: Find out how fast you have to throw a ball up into the air so that it will fall back to earth in exactly 100 seconds. The ball's motion—rising fast, slowing down, stopping, and coming back down—is exactly the right motion to make the time the maximum on a wristwatch strapped to the ball.

Now consider a slightly different game. We have two points *A* and *B* both on the earth's surface at some distance from one another. We play the same game that we did earlier to find what we call the straight line. We ask how we should go from *A* to *B* so that the time on our moving watch will be the longest—assuming we start at *A* on a given signal and arrive at *B* on another signal at *B* which we will say is 100 seconds later by a fixed clock. Now you say, "Well, we found out before that the thing to do is to coast along